

Home Lab Project
Week2, Day 3

Task: Create and prioritize the project backlog and implementation tasks.

Elaboration:

Develop a prioritized list of the tasks required to build, configure, test, and document the cybersecurity home lab. Organize the work in a logical sequence so that foundational components are completed before security testing begins.

The backlog should include the major implementation activities, such as:

- Prepare the computer and virtualization environment.
- Install and configure **Oracle VirtualBox**.
- Create and configure the **Kali Linux** VM.
- Create and configure the **Metasploitable2** VM.
- Install and configure **pfSense** as the virtual firewall.
- Configure the **Internal LAN and DMZ** network segments.
- Connect the virtual machines to the appropriate network segments.
- Verify network connectivity and firewall rules.
- Install/configure **Wireshark** for packet capture and analysis.
- Perform network reconnaissance and vulnerability assessment.
- Conduct controlled security testing against Metasploitable2.
- Capture and analyze network traffic with Wireshark.
- Document findings, configurations, and security observations.
- Test the completed lab and prepare final documentation.

Prioritize the backlog according to **dependencies, importance, and project sequence**. Tasks required to establish the lab environment should be completed before testing and analysis activities. Critical tasks should receive higher priority, while documentation and refinement tasks can follow the successful implementation and testing of the core environment.